

Reviewer Pack — Minimal Order of Noncommutative Integral Points and Frege Pr...

Entry fa1c8602fb55 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Noncommutative Integral Points and Frege Proof Depth Inequality

Entry ID: fa1c8602fb55 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-06 23:29:40 UTC

1. Conjecture

Field A (mathematical branch): Noncommutative Geometry (Integral Points) **Field B** (complexity object): Boolean Satisfiability (Frege Proof Complexity)

Statement:

For every boolean satisfiability instance with n variables, the minimal order of noncommutative integral points in the ring of integers of a finite field extension associated with the instance is linearly correlated with its Frege proof depth, such that $\text{MinimalOrder}(I) = \Theta(\text{FregeProofDepth}(n))$ for some integer constant I .

Rationale (proposer's reasoning):

Noncommutative geometry has shown potential in capturing algebraic structures underlying physical systems. The minimal order of noncommutative integral points could provide a new perspective on the complexity of proving satisfiability, potentially revealing deeper connections between algebra and computational problems.

Taxonomy category: NONCOMMUTATIVE_GEOMETRY_FREGEPROOF (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 5d84693b8fdb49b7

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for at least 80% of the $n \leq 40$ boolean satisfiability instances, $\text{MinimalOrder}(I)$ is within 30% of $\text{FregeProofDepth}(n)$, as determined by a correlation coefficient ≥ 0.7 over an aggregate statistic from multiple seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	0.80	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 6 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - intitle:Minimal Order Noncommutative Geometry AND Frege Proof Complexity - title:Frege Proof Depth AND noncommutative integral points boolean satisfiability - arxiv:math.NT OR cs.LO intitle:finite field extension Frege proof complexity

Top relevant hits considered: - [http://arxiv.org/abs/math/0610043v5] Lecture Notes on Noncommutative Algebraic Geometry and Noncommutative Tori - [http://arxiv.org/abs/1209.3595v2] Noncommutative complex differential geometry - [http://arxiv.org/abs/1612.06688v2] The Ricci Curvature in Noncommutative Geometry - [http://arxiv.org/abs/2402.10351v3] Quantum Automating $\mathbf{TC}^0$ -Frege Is LWE-Hard - [http://arxiv.org/abs/1311.2501v4] A reduction of proof complexity to computational complexity for $AC^0[p]$ Frege systems - [http://arxiv.org/abs/1409.5893v2] Fast evaluation of far-field signals for time-domain wave propagation

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.5s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_instance(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def frege_proof_depth(instance):
        # Placeholder function to simulate Frege proof depth
        # This is a dummy implementation and should be replaced with actual logic
        return len(instance)

    def minimal_order_of_noncommutative_integral_points(n):
        # Placeholder function to simulate the minimal order of noncommutative integral points
        # This is a dummy implementation and should be replaced with actual logic
        return n

    n = random.randint(5, 40)
    instance = generate_boolean_instance(n)
    frege_depth = frege_proof_depth(instance)
    minimal_order = minimal_order_of_noncommutative_integral_points(n)

    return {
        "metric_name": "MinimalOrder",
```

```

        "metric_value": minimal_order,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": abs(minimal_order - frege_depth) / max(1, frege_depth) <= 0.3,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] if len(sys.argv) > 1 else [
        2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71,
        73, 79, 83, 89, 97, 101, 103, 107, 109, 113
    ]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_metric_value = sum(r["metric_value"] for r in results) / len(results)
    std_metric_value = math.sqrt(sum((r["metric_value"] - mean_metric_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_metric_value} std={std_metric_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means it was unable to complete its execution and provide a result. | next: Re-run the test with increased time limits or optimize the code to ensure it completes within the given time frame.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13435
2	propose	ollama_remote	glm4:latest	0	0	12345
3	propose	ollama_remote	glm4:latest	0	0	12607
4	preregistration	ollama_remote	glm4:latest	0	0	9668
5	novelty	ollama_remote	glm4:latest	0	0	10381
6	novelty	ollama_remote	glm4:latest	0	0	9646
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	20079
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8145
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12415
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7894
11	judge	ollama_remote	glm4:latest	0	0	19655

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 136270 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/falc8602fb55.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/falc8602fb55.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/falc8602fb55.tar.gz` (if generated)